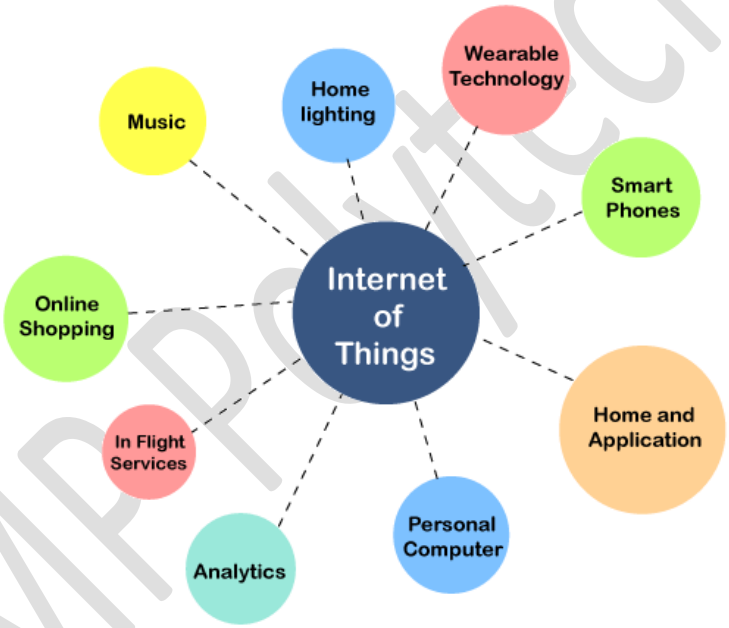
# Unit-I
# Introduction to IoT

## 1.1 IoT Definition

- The **Internet of Things (IoT)** provides the ability to interconnect computing devices, mechanical machines, objects, animals or unique identifiers and people to transfer data across a network without the need for human-to-human or human-to-computer is a system of conversation.
- **IoT applications** bring a lot of value in our lives.
- The Internet of Things provides objects, **computing devices**, or **unique identifiers** and people's ability to transfer data across a network without the **human - to human** or **human-to-computer interaction**.




## 1.2 IoT Characteristics

The way we define and demonstrate IoT is often explained by the characteristics of the Internet of Things (IoT). Here is a comprehensive list of all the significant characteristics of IoT:

1. **Connectivity**

- One of the most crucial characteristics of IoT infrastructure is connectivity.
- IoT things must stay connected to the infrastructure of IoT.
- There should be an all-time assurance that anyone can connect anytime, anywhere they want.
- For instance, the connection between internet gadgets like sensors and routers and the connection between one person and another via internet devices like smartphones.

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2. **Interoperable Communication Protocols**

- Seamless, interoperable communication is one of the key features of the Internet of Things.
- With the rising popularity of IoT, these IoT devices are now more interconnected than before. This is one of the most important characteristics of IoT
- This characteristics of IoT ensures that's the communication with one another work more seamlessly and efficiently.
- These devices/gadgets are connected to the world of the internet by leveraging cloud service.
- This gives room for seamless communication between devices.

3. **Identity of Things**

- It's a well-known fact that anyone's unique characteristic is their identity.
- For instance, all smartphones come with a unique IMEI number. This is true for all other smart devices on the planet.
- Identifying the things on the internet and people becomes easier with this unique number. Thus, it's safe to say the Identity of Things is among the most crucial characteristics of IoT.

4. **Scalability**

- Considering scalability in the system design has become more crucial than ever due to the surging ubiquity of the IoT.
- In the technical world, scalability refers to a system's capability to grow without adversely impacting its performance.
- Today, scalability is among the most popular characteristics of IoT.
- With the help of this characteristics of IoT, businesses can leverage the power of IoT by scaling it to fit their business.
- For instance, IoT can help hotel businesses track the rooms and amenities used by different guests.
- It also assists SMEs by sending targeted messages and offers on their behalf to customers.

5. **Dynamic or Self-Adapting**

- The dynamic nature is another key characteristic of IoT as it requires one to become self-adaptive to comprehend the modifications and upgrades happening around it and respond accordingly.
- For instance, the camera device was originally introduced to the world to take pictures.
- Today, the camera comes with many features that let you adjust pictures based on the light.
- Thus, being dynamic and self-adaptive is among the crucial characteristics of IoT that one cannot avoid.

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## 1.3 IoT Applications

**1. Wearables**

- Wearable technology is the hallmark of IoT applications and one of the earliest industries to deploy IoT.
- We have fit bits, heart rate monitors and smartwatches these days.
- Guardian glucose monitoring device has been developed to help people with diabetes.
- It detects glucose levels in our body, uses a small electrode called the glucose sensor under the skin, and relates it to a radiofrequency monitoring device.

**2. Smart Home Applications**

- The smart home is probably the first thing when we talk about the IoT application.
- The example we see the AI home automation**.** Where a string of musical notes uses in-house functions.

**3. Health care**

- IoT applications can transform reactive medical-based systems into active wellness-based systems.
- Resources that are used in current medical research lack important real-world information.
- It uses controlled environments, leftover data, and volunteers for clinical trials.
- The **Internet of Things** improves the device's **power, precision** and **availability**.
- IoT focuses on building systems rather than just tools.

**4. Smart Cities**

- Most of you have heard about the term smart city.
- Smart city uses technology to provide services.
- The smart city includes improving transportation and social services, promoting stability and giving voice to their citizens.

## 1.4 Key Components of IoT System Things/Device, Gateway, Cloud/Server, Analytics, User Interface

1. **Things or Device**
   These are fitted with sensors and actuators. Sensors collect data from the environment and give to gateway where as actuators performs the action (as directed after processing of data).

2. **Gateway**
   The sensors give data to Gateway and here some kind of pre-processing of data is even done. It also acts as a level of security for the network and for the transmitted data.

3. **Cloud**
   The data after being collected is uploaded to cloud. Cloud in simple terms is basically a

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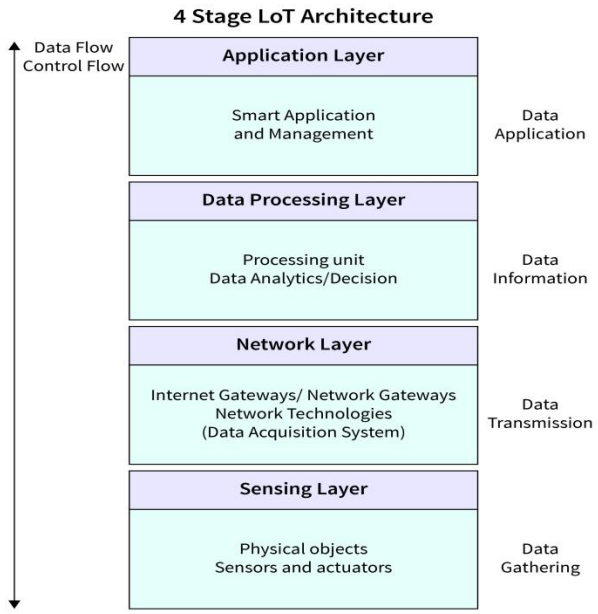
set of servers connected to internet 24*7.

4. **Analytics**
   The data after being received in the cloud processing is done . Various algorithms are applied here for proper analysis of data (techniques like Machine Learning etc are even applied).

5. **User Interface**
   User end application where user can monitor or control the data.

## 1.5 Architecture of IoT




- **Sensing Layer** − The first stage of IoT includes sensors, devices, actuators etc. which collect data from the physical environment, processes it and then sends it over the network.
- **Network Layer** − The second stage of the IoT consists of Network Gateways and Data Acquisition Systems. DAS converts the analogue data (collected from Sensors) into Digital Data. It also performs malware detection and data management.
- **Data Processing Layer** − The third stage of IoT is the most important stage. Here, data is pre-processed on its variety and separated accordingly. After this, it is sent to Data Centres. Here Edge IT comes into use.
- **Application Layer** − The fourth stage of IoT consists of Cloud/Data Centres where data is managed and used by applications like agriculture, defence, health care etc.

## 1.6 IoT Challenges

### 1.6.1 Design Challenges (Connectivity, Power Requirements, Complexity, Storage and Computational Capability, Data Extraction from complex environment)

1. **Interoperability:**
   - IoT involves a multitude of devices from different manufacturers, operating on various platforms and communication protocols. Ensuring seamless interoperability and communication among these diverse devices can be challenging.

2. **Security and Privacy:**
   - IoT devices often handle sensitive data, and security vulnerabilities can lead to privacy breaches. Designing robust security measures, including encryption, secure authentication, and regular software updates, is crucial to safeguarding user data and maintaining trust.

3. **Scalability:**
   - IoT ecosystems can rapidly grow in scale as more devices are added. Designing systems that can scale efficiently to accommodate a large number of devices, data points, and users is a significant challenge.

4. **Power Consumption and Efficiency:**
   - Many IoT devices operate on limited power sources, such as batteries. Balancing functionality with low power consumption is critical for prolonging device lifespans and reducing the need for frequent battery replacements.

5. **Data Management and Analytics:**
   - IoT devices generate vast amounts of data. Designing effective strategies for data collection, storage, and analysis is essential. This includes addressing issues related to data volume, real-time processing, and determining what data should be sent to the cloud for analysis.

### 1.6.2 Security Challenges (Security & Personal Safety, Privacy)

1. **Device Authentication and Authorization:**

   - Verifying the identity of devices and ensuring that only authorized devices can access and interact with the network is a significant challenge. Weak authentication mechanisms can lead to unauthorized access and potential security breaches.

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2. **Data Encryption:**
    - Transmitting sensitive data between IoT devices and networks requires robust encryption. Weak or improperly implemented encryption methods can expose data to interception and unauthorized access.
3. **Insecure Interfaces and APIs:**
    - Inadequately secured user interfaces and application programming interfaces (APIs) can become entry points for attackers. Properly securing interfaces and APIs is crucial to preventing unauthorized access and manipulation of IoT devices.
4. **Lack of Standardization:**
    - The absence of standardized security protocols across IoT devices can create vulnerabilities. A lack of uniform security practices makes it challenging to implement consistent security measures across diverse IoT ecosystems.
5. **Physical Security:**
    - Physical tampering with IoT devices can compromise their security. Designing devices with robust physical security features and implementing measures to detect and respond to physical attacks is essential.

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